Subashanan Nair

Data Scientist

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Professional Summary

Data Scientist with 7+ years of experience across machine learning, data engineering and technical education. Builds production data pipelines and evidence based tools for public and organisational datasets. Recent work focuses on Malaysian public statistics: a reproducible Penang geography study, leakage safe traffic forecasting and a benchmark of whether LLMs can find the latest official figure when open data is stale. Hands on with Python, scikit-learn, XGBoost, LightGBM, Streamlit, Docker and GitHub Actions. DataCamp certified AI Engineer for Data Scientists and Kaggle Top 100 competitor.

Skills

- **Languages:** Python, SQL, R, JavaScript, TypeScript
- **ML and AI:** scikit-learn, XGBoost, LightGBM, CatBoost, deep learning, reinforcement learning, LLM applications (Ollama, prompt design, RAG, text to SQL)
- **Data and MLOps:** Pandas, NumPy, MLflow, Docker, GitHub Actions, Streamlit, Plotly, Tableau, PostgreSQL, FastAPI
- **Web:** Next.js, FastAPI, REST API design, Supabase, Quarto

Experience

Forward College | Instructor and Data Scientist *Feb 2019 to Present*

- Instruct 3 cohorts per year in Python, machine learning and data analytics, averaging a 4.7/5 learner satisfaction score.
- Built internal tools: an LLM assisted project review dashboard, a GitHub based submission and assessment platform and an exam monitoring app.
- Engineered and deployed ML pipelines and predictive models with Python, scikit-learn and MLflow, reducing manual analysis time by 60%.
- Delivered data solutions and dashboards for 3+ organisations, including Penang Institute.

Penang Institute | Data Scientist (Consultant) *Jun 2024 to Present*

- Diagnosed a stale public statistics site and replaced a 34 script R workflow with an automated Python pipeline (GitHub Actions, Docker) publishing OpenDOSM and data.gov.my to Google Drive for Tableau.
- Built a weekly ingestion pipeline for key Penang indicators (population, GDP, employment, CPI, household income) backed by a 31 test validation suite.
- Created a pull request gated Streamlit uploader so staff can add new publications without touching code,

and an interactive key statistics dashboard (Streamlit, Plotly) on Streamlit Cloud.

- Ran the website revamp and maintain the Quarto statistics website.
- **Related published studies:** Penang Pulse (equity of access across districts), Forecasting Lab (traffic forecasting), LLM Eval Lab (freshness of open data) and Penang AI Assistant evaluation.

Kuen Lai International Secondary School | STEAM Consultant *May 2013 to Jan 2019*

- Taught Python programming, Arduino and Raspberry Pi electronics and 3D printing through hands on project based STEAM lessons.
- Designed STEAM programmes aligned with national and international curriculum frameworks.
- Ran workshops for educators on STEAM pedagogy and digital literacy.

Published Projects

All figures below come from verified runs. Source reports carry commit hashes and checksums.

Penang Pulse | Spatial Analysis

Python, NetworkX, OpenStreetMap, WorldPop, DOSM data, Next.js

- Reproducible study of how evenly public clinics, schools and transit stops are distributed across Penang districts, using road distances on the OpenStreetMap network.
- Result: 20% of Seberang Perai Selatan residents live more than 3 km by road from a public clinic; 74% of Penang residents live within 1 km by road of a Rapid Penang stop; 2,437 public clinics, schools and stops mapped.
- Two independent population maps agree on every district; SHA-256 checksums and reproducible commands are published with the study.

Forecasting Lab | Predictive Modelling

Python, scikit-learn, SciPy, PyTorch, Chronos, UCI dataset 608

- Leakage safe 15 minute traffic forecasting across a 36 sensor network, with gradient boosting and a zero-shot foundation model (Chronos).
- Result: gradient boosting cut MAE by 23% versus persistence at 15 minutes (95% interval 16 to 30%); zero-shot Chronos beat gradient boosting by 26% at 2 hours (interval 17 to 34%); Chronos had the lowest error at every horizon.
- Five models compared on a held back test window, with bootstrap intervals.

LLM Eval Lab | Published

Python, Ollama Cloud (gpt-oss:120b, deepseek-v4.1-flash), bootstrap intervals

- Benchmark of whether cloud models can find the latest official Malaysian figure when open data lags the release.
- Result: across 120 hand verified questions, gpt-oss:120b answered 12 of 20 latest questions correctly with tools versus 1 without; deepseek-v4.1-flash got 12 of 20 with tools versus 0 without.
- Question set and reproducible run under version control, with snapshot checksums.

Penang Economic Data Assistant | Prototype and Evaluated

FastAPI, PostgreSQL, Next.js, Docker, Ollama Cloud (gpt-oss:120b)

- Natural language assistant over OpenDOSM CPI and unemployment data; questions become read-only SQL against approved views and are recorded in an audit log.
- Evaluation: 6 of 6 passing gates, 81 of 81 correct supported answers, 45 of 45 critical scope cases, p50 1.6 s latency. Provider comparison against Gemini shows near identical accuracy.

Other Projects

- **Documentation MCP Server** (Node.js, Docker): aggregates live docs for React, Vue, Angular and more into a searchable REST API with daily cron updates.
- **FC Submission Platform** (Next.js, TypeScript, Supabase, GitHub OAuth, Playwright): deadline snapshotting of student repos with automated deliverable checks and instructor override trail.
- **Exam Guard** (Rust, Tauri): LAN proctoring tool needing no accounts or internet; cross platform Windows, macOS and Linux releases.

Experience (Internal Tools)

- **Forward College project review tool:** monitors 9 student capstone repositories and scores each against a 10 section rubric. The LLM reviewer must cite exact source quotes; fabricated quotes are rejected and the final score is computed locally, not taken from the model. Runs hourly on GitHub Actions; never executes student code.

Education

- **Asia e University:** Master of Education in Science, May 2014 to Oct 2016
- **Universiti Tun Abdul Razak:** Bachelor of Education (Hons), May 2009 to Sep 2012

Certifications

- DataCamp | AI Engineer for Data Scientists Associate, Apr 2026 (Certificate ID: AEDS0 019 212 905 886)
- Coursera (DeepLearning.AI and Stanford) | Supervised ML: Regression and Classification, 2024
- Google for Education | Google Certified Educator Level 1, Dec 2021
- Kaggle Competition | Top 100 placement in a global machine learning competition